

PROMPTMINER-SAM2: PROMPT MINING AND UNCERTAINTY-GUIDED REFINEMENT FOR REMOTE SENSING INSTANCE SEGMENTATION

Cheng Wang^{1,2}, Baotian Shi², Xiaoyu Li¹, Han Zhang^{2*}

¹School of Cybersecurity, National School of Secrecy

²School of Artificial Intelligence, Optics and Electronics (iOPEN)
Northwestern Polytechnical University, Xi'an, China

ABSTRACT

Accurate remote-sensing instance segmentation remains challenging in complex scenes. Existing methods decode masks from latent candidate representations, without explicitly using shape cues for irregular or adjacent boundaries. To address this limitation, we propose PromptMiner-SAM2, an end-to-end framework for remote-sensing instance segmentation. It predicts per-candidate coarse shapes first and mines the resulting shape cues for mask decoding. Specifically, the coarse shape prompt mining module (CSPM) predicts the coarse shapes from candidate region-level features and generates shape-aware prompts accordingly. Moreover, a decoder-tail module (UDR) is designed to identify uncertain pixels and correct the predictions using local mask features and instance context. Experiments on the WHU and NWPU VHR-10 benchmarks show that PromptMiner-SAM2 achieves mask AP values of 73.4% and 68.3%, respectively, attaining the highest mask AP among the compared methods.

Index Terms— Instance segmentation, remote sensing, SAM2, prompt mining, mask refinement

1. INTRODUCTION

Remote-sensing instance segmentation assigns each object a pixel-level mask, supporting urban planning and disaster assessment [1,2]. High-resolution remote-sensing images often exhibit scale variation and complex background textures. These factors complicate accurate instance-mask prediction.

Existing instance segmentation methods use two-stage, one-stage, conditional-mask, or set-prediction paradigms and decode masks from region features, dynamic parameters, or instance queries [3,4,5,6,7]. In remote-sensing imagery, prior work improves mask prediction through multi-scale feature fusion, boundary modeling, and foundation-model adaptation [1,2,9,15]. SAM series models enable prompt-conditioned mask decoding [10,11], while related work explores learned and adapted prompts [12,13,16].

However, many approaches decode masks from recognition and localization features without explicit candidate-local

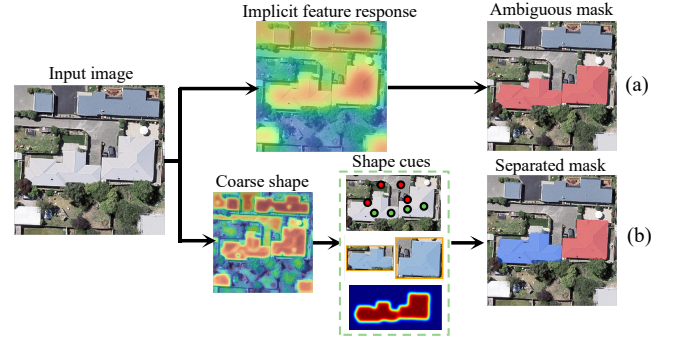


Fig. 1. Effect of coarse-shape prompt mining. (a) An implicit feature response produces an ambiguous mask for adjacent instances. (b) A coarse shape yields shape cues that form diverse prompts and separate the instances.

shape cues. The decoder receives limited shape guidance for irregular contours or adjacent instances. Prior work explores learned spatial prompts for automatic SAM segmentation [12, 21] and prompt-based mask refinement [14,22]. These studies suggest that explicit shape cues can provide complementary guidance for instance-mask decoding. For this purpose, we introduce such explicit shape cues into end-to-end remote-sensing instance segmentation, yield better performance on this task, as illustrated in Fig. 1. To this end, we propose PromptMiner-SAM2.

Our main contributions are summarized as follows:

- We propose **PromptMiner-SAM2**, a two-stage framework that mines shape cues from per-candidate coarse shapes to form explicit SAM2 prompts for end-to-end remote-sensing mask prediction.
- We introduce **CSPM** and **UDR**: CSPM predicts a per-candidate coarse shape and mines diverse prompts from it, while UDR selectively corrects low-confidence predictions to improve mask quality.
- We conduct extensive experiments on the WHU and NWPU VHR-10 benchmarks, showing the effectiveness of explicit shape cues for mask prediction and uncertainty-driven refinement.

*Corresponding author: Han Zhang(zhanghan9937@nwpu.edu.cn).

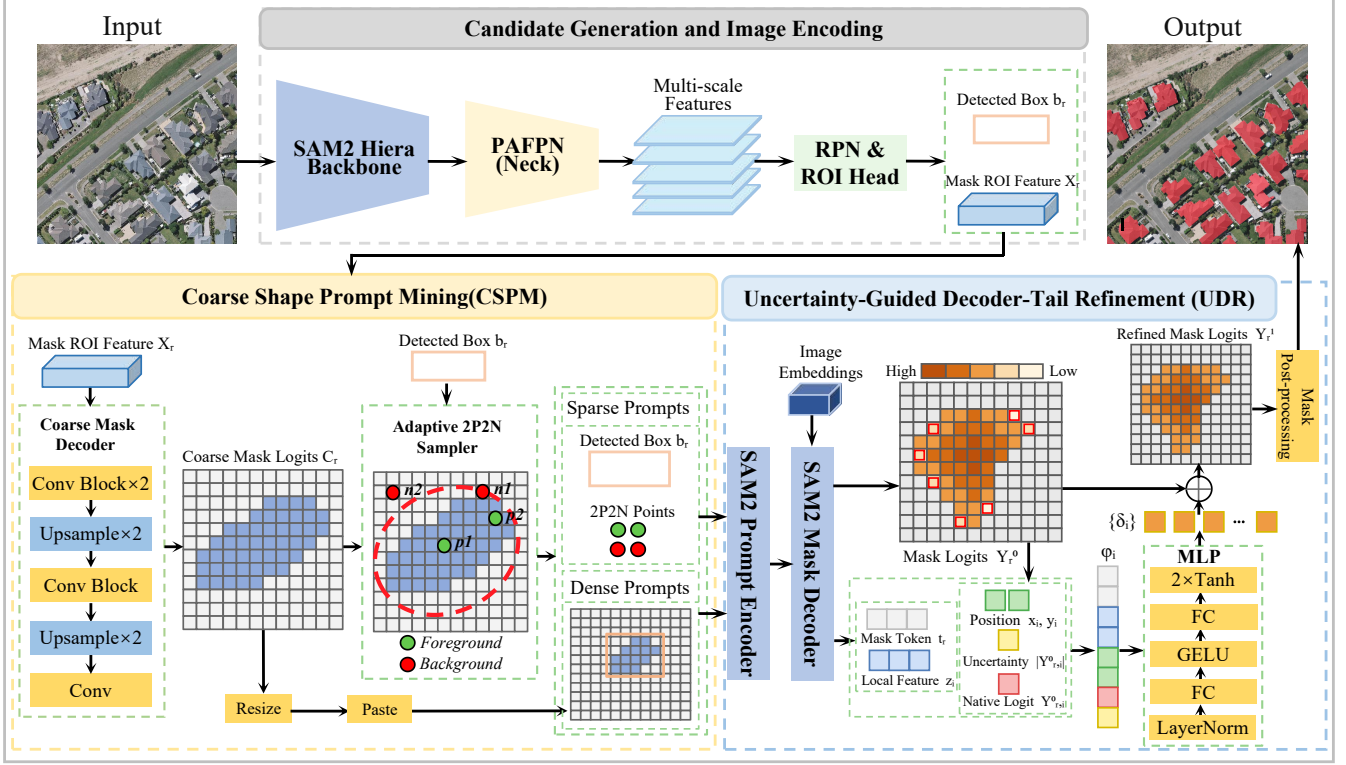


Fig. 2. Overview of the proposed PromptMiner-SAM2. The feature-extraction branch produces candidates, CSPM provides SAM2 with coarse-shape-derived prompts, and UDR corrects uncertain mask pixels.

2. METHOD

2.1. Overall Architecture

Fig. 2 illustrates the overall architecture of the proposed PromptMiner-SAM2 for remote-sensing instance segmentation, which couples a two-stage detector with the SAM2 prompt-based mask decoder and associates instance localization to mask prediction in one process. Given an image I , the Hiera encoder of SAM2 extracts multi-scale features. The detection branch produces candidate regions and RoI features through the PAFPN, RPN, and RoI pathways. The same Hiera features serve as image embeddings for subsequent SAM2 mask decoding. For each candidate, the proposed CSPM predicts a per-candidate coarse shape and mines sparse and dense prompts from it. SAM2 decodes these prompts with the image embedding into an initial instance mask, and UDR corrects the uncertain pixels of that mask.

2.2. Coarse-Shape Prompt Mining

Let r denote a candidate region with RoI feature X_r and detection box b_r , which defines the spatial extent and local coordinate reference. The proposed CSPM predicts a candidate-local coarse shape in this box-local frame and uses it to generate complementary sparse and dense prompts.

The coarse-shape branch predicts C_r :

$$C_r = f_{\text{coarse}}(X_r), \quad C_r \in \mathbb{R}^{1 \times 64 \times 64}. \quad (1)$$

During training, the ground-truth mask is cropped by the candidate box and represented in the same frame to supervise C_r . Auxiliary supervision guides C_r to capture the foreground layout before prompt generation.

The foreground probability map is $p_r = \sigma(C_r)$, and $\bar{d}_r(u)$ denotes the normalized internal distance from u to the predicted foreground boundary. For each candidate, point prompts comprise two positive and two negative points (2P2N) mined from the coarse shape:

$$\begin{aligned} u_i^p &= \arg\max_{u \in \Phi_p} p_r(u) \bar{d}_r(u), \\ u_i^n &= \arg\max_{u \in \Phi_n} (1 - p_r(u)), \end{aligned} \quad i = 1, 2, \quad (2)$$

where Φ_p contains confident pixels well inside the predicted foreground support and Φ_n contains low-probability pixels just outside the support and in its surrounding background. Points are selected sequentially, and each selected point excludes a small neighborhood of the previously selected ones, so the two points of each polarity cover complementary parts of the instance.

The box b_r is the common spatial reference for all three prompt types. The selected points are mapped to image coordinates and, with their positive or negative labels, form the

sparse prompt set $\mathcal{S}_r$. The coarse-shape logits are resized and placed into the image region covered by b_r , forming the dense prompt Q_r , while b_r itself serves as the box prompt. SAM2 decodes the three prompts jointly with the image embedding:

$$Y_r^0 = \mathcal{D}(\mathcal{E}_I(I), \tilde{\mathcal{E}}_P(\mathcal{S}_r, b_r, Q_r)), \quad (3)$$

where $\mathcal{E}_I$, $\tilde{\mathcal{E}}_P$, and $\mathcal{D}$ denote the image encoder, the prompt-embedding pathway, and the mask decoder. Q_r keeps the signed coarse-shape logits and carries continuous foreground/background confidence into decoding.

2.3. Uncertainty-Guided Decoder-Tail Refinement

Although CPM gives the decoder explicit conditions derived from the coarse shapes, the initial masks may still contain locally uncertain predictions. A single decoding pass produces the initial logit map Y_r^0 , the aligned decoder feature map U_r , and the mask token t_r . UDR predicts bounded residual corrections at a few selected low-confidence mask pixels and leaves all other logits unchanged.

UDR identifies mask pixels with the least decisive predictions on the decoder output grid $\mathcal{G}_r$. Smaller-magnitude native logits indicate less decisive decoder states, so UDR keeps the K_r pixels of smallest magnitude

$$\begin{aligned} K_r &= \min(K, |\mathcal{G}_r|), \\ \Omega_r &= \text{BottomK}(|Y_r^0|, K_r), \quad K = 64, \end{aligned} \quad (4)$$

BottomK returns the indices of the K_r smallest entries of $|Y_r^0|$, which identify the mask pixels kept for refinement.

For each selected mask pixel $i \in \Omega_r$, UDR samples the local decoder feature z_i from U_r and concatenates it with the mask token, the normalized coordinates, and the current logit:

$$\phi_i = [z_i, t_r, Y_{r,i}^0, x_i, y_i, |Y_{r,i}^0|]. \quad (5)$$

Here, x_i and y_i are the normalized coordinates of i in $\mathcal{G}_r$. Thus, ϕ_i couples local evidence, instance context, and the current prediction state. A lightweight MLP g then predicts a bounded residual correction:

$$\delta_i = 2 \tanh g(\phi_i), \quad i \in \Omega_r, \quad (6)$$

Finally, the residuals are written back:

$$Y_{r,i}^1 = \begin{cases} Y_{r,i}^0 + \delta_i, & i \in \Omega_r, \\ Y_{r,i}^0, & i \notin \Omega_r. \end{cases} \quad (7)$$

This update modifies only the selected low-confidence mask pixels. At each selected pixel, a positive/negative-balanced one-sided margin loss uses the deployment boundary $b = \log(0.4/0.6)$ and margin $m = 0.10$, vanishing once the refined logit crosses $b \pm m$ on the ground-truth side. UDR provides localized correction after shape-conditioned decoding.

Table 1. Sensitivity of K on the NWPU VHR-10 dataset under a common development protocol.

K	Bbox (%)			Mask (%)		
	AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅
32	69.38	91.69	79.03	64.94	92.91	70.01
64	70.35	92.23	81.65	66.65	94.15	71.97
128	70.13	92.42	81.32	65.97	93.71	72.22

3. EXPERIMENTS

3.1. Experimental Setup

We evaluate the method on two remote-sensing benchmarks: WHU and NWPU VHR-10. The WHU dataset is a single-class building instance segmentation benchmark with 2,943/627/2,220 training, validation, and test images [17]. The NWPU VHR-10 dataset is a ten-class object detection benchmark [18]. As the original release provides boxes only, we use the instance masks and 520/130 split of Su *et al.* [19].

The model is implemented in PyTorch and trained on four NVIDIA RTX 3090 GPUs. Each GPU processes one image, with two gradient-accumulation steps yielding an effective batch size of eight. Both benchmarks are initialized from SAM2 Hiera-Base+ weights, with the Hiera backbone frozen and the adaptation modules, prompt pathway, and mask decoder trained. Training uses AdamW with an initial learning rate of 5×10^{-4} and cosine annealing. The objective joins the detection, coarse-shape, and final instance-mask losses.

Table 1 compares UDR under $K \in \{32, 64, 128\}$ in a common development protocol. The main experiments use $K = 64$, which gives the highest overall bbox and mask AP.

3.2. Comparative Analysis

The proposed PromptMiner-SAM2 is compared with 10 representative instance-segmentation methods. Table 2 groups them into CNN-based methods (CNN) and global sequence modeling methods (GSM). CNN includes two-stage, one-stage, and cascade designs, whereas GSM includes Transformer-style mask prediction and foundation-model prompting.

It can be concluded from Table 2 that our method obtains the highest mask AP on both datasets, by 0.1 and 2.2 percentage points over the strongest competitors. On the WHU dataset, it also gives the best mask AP₇₅, while its bbox AP is 76.4% versus 76.7% for YOLO11s-seg. On the NWPU VHR-10 dataset, it attains the best bbox AP, bbox AP₇₅, mask AP, and mask AP₅₀. Although its mask AP₇₅ trails the best overall result, it exceeds every GSM method.

Figure 3 shows qualitative results. Compared with other methods, our method separates touching building roofs on the WHU dataset and reduces erroneous merging of adjacent instances on the NWPU VHR-10 dataset.

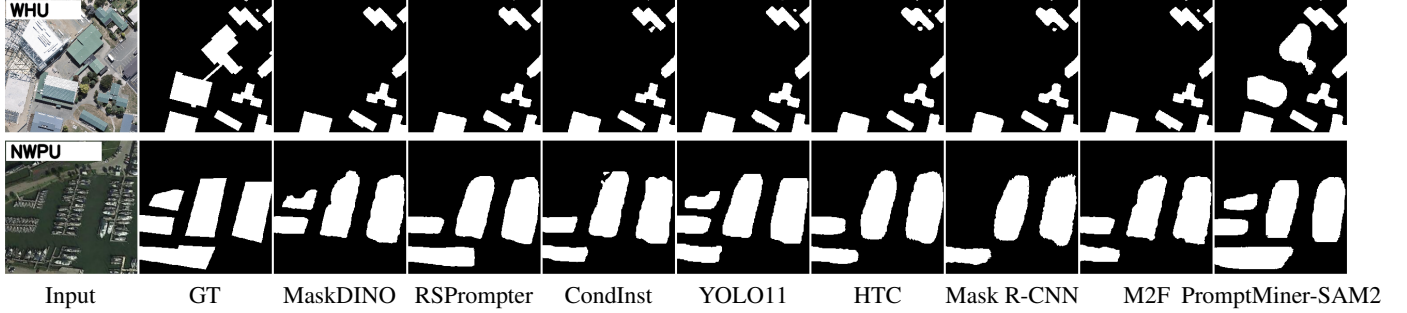


Fig. 3. Qualitative comparison of existing methods and PromptMiner-SAM2 on the WHU and NWPU VHR-10 datasets.

Table 2. Comparative analysis on the WHU and NWPU VHR-10 datasets. Table reports AP / AP₅₀ / AP₇₅ for Bbox and Mask.

Type	Method	WHU						NWPU VHR-10					
		Bbox (%)			Mask (%)			Bbox (%)			Mask (%)		
		AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅
CNN	Mask R-CNN [3]	70.9	89.9	80.9	68.2	90.0	80.0	65.4	90.2	78.9	63.0	90.1	71.2
	MS R-CNN [4]	68.8	89.4	80.0	67.7	89.6	79.5	67.8	94.3	83.8	64.7	93.6	80.8
	HTC [5]	72.7	90.6	82.5	69.2	90.7	81.6	58.9	86.9	67.7	51.3	84.6	53.3
	CondInst [6]	74.2	90.0	82.4	69.7	90.2	80.9	65.0	90.1	79.6	62.8	90.7	68.1
	YOLO11s-seg [20]	76.7	91.2	85.4	66.8	91.3	79.1	71.0	94.4	79.5	63.9	93.9	67.3
GSM	Mask2Former [7]	61.4	82.7	70.7	64.5	84.1	75.6	60.3	77.9	66.8	61.5	84.9	67.8
	MaskDINO [8]	73.3	90.8	82.7	73.3	90.3	82.5	65.7	88.9	74.2	65.8	89.6	71.5
	RS4D-box [9]	64.5	87.6	74.7	64.1	87.8	75.2	57.0	82.5	65.4	54.3	80.1	57.6
	RSIISN [1]	74.1	90.0	84.4	72.1	90.1	83.9	66.7	91.5	78.4	62.9	89.7	67.1
	RSPrompter [12]	72.5	91.0	81.7	72.5	92.0	82.9	70.3	93.6	81.0	66.1	92.7	70.6
Ours	PromptMiner-SAM2	76.4	91.5	85.4	73.4	91.4	85.5	73.4	94.0	84.2	68.3	94.5	74.6

Table 3. Prompt-path and UDR module ablation on the NWPU VHR-10 dataset, where P , B , and M denote point, box, and dense shape prompts, respectively, and UDR is the uncertainty-guided decoder-tail refiner.

Variant	Bbox (%)			Mask (%)		
	AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅
P	71.81	93.60	81.89	63.15	92.56	67.73
B	72.14	94.34	82.24	65.25	94.12	69.02
$P+B$	71.74	93.52	81.68	66.94	93.93	72.73
$P+B+M$	73.76	93.80	84.69	66.35	93.66	71.61
$P+B+M + \text{UDR}$	73.43	93.96	84.16	68.28	94.53	74.61

3.3. Ablation Study

Point-only P serves as the baseline in Table 3. Adding box prompt B raises mask AP/AP₅₀ from 63.15%/92.56% to 65.25%/94.12%. The box specifies the candidate extent, and the gain indicates more reliable foreground-background separation. Combining $P+B$ reaches 66.94% mask AP and 72.73% AP₇₅ while bbox AP remains unchanged. This result suggests that the two prompt types provide complementary

mask-decoding constraints. After M is added, bbox AP/AP₇₅ rise from 71.74%/81.68% to 73.76%/84.69%, but mask AP/AP₇₅ are 66.35%/71.61%. This contrast indicates that the coarse shape supports region-level localization while boundary ambiguity remains. UDR then raises mask AP/AP₇₅ to 68.28%/74.61% with comparable bbox AP. It updates only low-confidence mask pixels using local decoder features and instance context. The higher mask scores with comparable bbox AP indicate that this targeted update improves mask delineation without materially affecting candidate localization.

4. CONCLUSION

In this work, we present PromptMiner-SAM2, which mines SAM2 prompts from coarse shapes and selectively refines uncertain mask pixels. Results on the WHU and NWPU VHR-10 datasets show strong, balanced detection and segmentation performance. In the future, it is a promising direction to explore prompts adaptive to object scale and scene context, particularly for improving cross-sensor transfer and delineating small or densely adjacent instances.

5. REFERENCES

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